

01

GlucoTwin AI

A Patient-Specific Digital Twin for Early Prediction of Glucose Spikes

“Simulate the Patient. Predict the Risk. Act Before the Spike.”

- Challenge: Happiest Health Digital Twin Challenge 2026
- Category: Chronic Disease Management & Preventive Healthcare
- Target: Type 2 Diabetes Postprandial Glucose Volatility
- Status: Functional Research & Decision-Support Prototype

The Problem: Acute Postprandial Spikes

Why Glycemic Spikes are the Silent Drivers of Diabetic Complications

HbA1c alone misses dangerous intraday glycemic excursions.

- Postprandial spikes (≥ 180 mg/dL) cause acute vascular endothelial shear stress.
- Independent driver of cardiovascular mortality, nephropathy, and diabetic retinopathy.
- Traditional care is reactive: finger-prick tests detect spikes only AFTER damage occurs.
- Clinical Need: An early-warning horizon (2 hours ahead) to intervene proactively.

Epidemic Scale: Why This Matters in India

Addressing the Diabetes Capital of the World

India has over 101 million individuals diagnosed with diabetes.

- Indian phenotype: High visceral adiposity and early beta-cell dysfunction at lower BMI.
- Dietary reality: High glycemic index carbohydrate staples (white rice, refined wheat).
- Doctor-patient ratio deficit (1:1,500+) requires automated decision-support.
- Scalable digital twins empower proactive lifestyle interventions before clinical escalation.

The Solution: GlucoTwin AI

Continuous Physiological State Simulation for Early Warning

Fusing long-term clinical history with high-frequency dynamic IoT telemetry.

- Continuously assimilates Static EHR + Continuous CGM + Wearable IoT.
- Predicts acute spike risk within a 2-hour forward horizon.
- Forecasts continuous multi-horizon glucose trajectories (+30, +60, +90, +120 min).
- Delivers SHAP-driven explainability and interactive counterfactual simulation.

What Makes It a True Digital Twin?

Beyond Simple Static Classification

A digital twin is a living, evolving virtual representation of a physical patient.

- Patient Profile: Fixed biological traits (age, BMI, genetics, baseline beta-cell reserve).
- Temporal Memory: Sliding 24-hour historical buffer capturing circadian rhythms.
- Current Metabolic State: Updated dynamically upon every 15-minute sensor observation.
- What-If Sandbox: Clones twin state to test counterfactual lifestyle interventions in silico.

Multi-Modal Data Fusion

Harmonizing Heterogeneous Healthcare Data Streams

Static clinical depth meets dynamic temporal resolution.

- Static Stream (Synthea EHR): HbA1c, fasting glucose, diabetes duration, medications.
- Dynamic Stream (IoT/CGM): 15-min interstitial glucose, HR, HRV, steps, sleep, meals.
- Zero Data Leakage: Strictly backward-looking sliding windows for rolling statistics.
- Cross-Modal Interactions: Activity-to-glucose ratio, sleep debt x glucose strain, carb balance.

System Architecture Overview

End-to-End Modern Microservice Architecture

Clean separation of ingestion, physics engine, ML pipeline, and doctor UI.

- Ingestion Layer: Temporal alignment, missing packet imputation, 15-min resampling.
- Digital Twin State Engine: In-memory patient twin management & memory buffering.
- ML Prediction Engine: XGBoost classifier, multi-output regressor, SHAP TreeExplainer.
- Presentation: Doctor analytics dashboard & What-If sandbox in React + Tailwind.

The Digital Twin State Engine

Continuous Biological State Evolution

Real-time state update cycle: Observe → Update → Re-feature → Predict → Alert.

- Class PatientDigitalTwin encapsulates clinical memory, state, and ML adapters.
- Maintains 96-observation sliding window (24 hours) for diurnal baseline tracking.
- Event-driven execution: New wearable packet triggers immediate state recalibration.
- State cloning enables zero-side-effect counterfactual sandbox simulations.

Machine Learning Pipeline

Rigorous Evaluation with Honest Benchmarks

Prioritizing interpretability, high sensitivity, and reliable generalization.

- Baseline Model: Logistic Regression (balanced weights, standardized features).
- Primary Model: XGBoost Classifier with 33 engineered physiological features.
- Comparator Benchmark: Random Forest Ensemble for non-linear comparison.
- Multi-Horizon Trajectory Regressor: MultiOutputRegressor forecasting +30m to +120m.

Both probabilistic risk score and tangible mg/dL forecasted curves.

- Binary Classification: P(Spike in next 2 hours) with clinical threshold ≥ 180 mg/dL.
- Risk Tiering: LOW ($<35\%$), MODERATE (35–64%), HIGH ($\geq 65\%$).
- Trajectory Projection: Point forecasts at +30m, +60m, +90m, +120m.
- Trend Dynamics: Categorizes rate of rise (Rapid Increase, Steady, Decreasing).

Explainable AI (SHAP)

Building Clinical Trust with Transparent Attribution

Doctors must understand WHY the twin predicts high risk.

- TreeExplainer computes exact additive feature attribution in milliseconds.
- Distinguishes Risk Amplifiers (e.g. high current glucose, carb load) from Protectors (steps).
- Physiological Translation: Maps raw feature values to human clinical mechanisms.
- Zero Hallucination: Explanations are mathematically derived directly from the model.

What-If Counterfactual Simulator

Simulate the Intervention Before Clinical Execution

Empowering doctors and patients with actionable lifestyle scenario modeling.

- Allows adjusting: Sleep duration (hrs), postprandial steps, meal carbs (g), glucose.
- Clones current patient state and re-computes physiological feature matrix.
- Side-by-side delta visualization: Current 78% Risk → Simulated 46% Risk.
- Mandatory Safety Notice: Clearly labeled as simulation, not prescription.

Clinical Doctor Dashboard

Built for High-Cognitive-Efficiency Healthcare Delivery

Modern, minimal, doctor-friendly interface eliminating cognitive overload.

- Patient Overview: Comprehensive EHR demographics, HbA1c, and prescription cards.
- Live Telemetry Timeline: Synchronized CGM, HR, HRV, steps, and meal markers.
- Prediction Gauge: Visual circular risk gauge with 2-hour trajectory reference bands.
- Interactive Sandbox: Live sliders for rapid clinical hypothesis testing.

Measured Model Performance

Evaluated on 10 Unseen Test Patients (6,640 Samples)

Actual empirical results with zero temporal leakage.

- XGBoost Primary: ROC-AUC = 0.9894 | PR-AUC = 0.9833 | F1 = 0.9217 | Recall = 90.99%
- Random Forest: ROC-AUC = 0.9831 | PR-AUC = 0.9752 | F1 = 0.9074 | Recall = 89.95%
- Logistic Regression: ROC-AUC = 0.9769 | PR-AUC = 0.9642 | F1 = 0.8858 | Recall = 90.39%
- Trajectory MAE: +30m: 7.50 mg/dL | +60m: 10.19 mg/dL | +90m: 12.67 mg/dL | +120m: 14.48 mg/dL

Demonstration Workflow

Three Distinct Evaluator Patient Personas

Deterministic clinical verification across distinct risk profiles.

- Patient PT-101 (Eleanor Vance): Low Risk, high activity (8.5k steps), HbA1c 6.4%.
- Patient PT-102 (Rajesh Kumar): Moderate Risk, erratic sleep (6.2h), HbA1c 7.6%.
- Patient PT-103 (Marcus Chen): High Risk, 95g carb lunch, sedentary, HbA1c 9.1%.
- Live Ingestion Replay: Auto-stream mode advances 15-min clock every 3 seconds.

Privacy, Security & Data Ethics

Patient Protection by Architectural Design

Zero identifiable patient data used in model development or deployment.

- 100% Synthetic Synthea-grounded EHR & physiologically simulated wearable telemetry.
- HIPAA & GDPR Compliant Architecture: No PII stored, logged, or exposed.
- Deterministic Seed Generation (Seed 42) ensures 100% audit reproducibility.
- Local runtime execution ensures no external data exfiltration.

System Limitations

Scientific Honesty & Technical Boundaries

Rigorous self-audit of prototype constraints.

- Sensor lag: Interstitial CGM readings lag capillary blood glucose by 5–15 minutes.
- Synthetic data domain gap: Real clinical data exhibits noise, artifacts, and dropout.
- Medication modeling: Does not account for pharmacokinetics of complex insulin titrations.
- Psychological stress: Cortisol spikes not directly measured by basic wrist wearables.

Future Scope & Road Ahead

From Prototype to Clinical Trial Readiness

Next-generation enhancements for GlucoTwin AI.

- Clinical Integration: FHIR (Fast Healthcare Interoperability Resources) HL7 API connector.
- Direct Sensor Ingestion: Dexcom Web API & Apple HealthKit / Google Health Connect sync.
- Closed-Loop Support: Automated micro-bolus insulin pump integration (Artificial Pancreas).
- Multi-Modal Vision: Smartphone meal photo carb estimation via on-device vision models.

Moving from reactive treatment to proactive digital twin prevention.

- Reduces Glycemic Variability (CV <36% target) and expands Time-In-Range (TIR >70%).
- Prevents Microvascular & Macrovascular Complications: Lowers long-term dialysis & ICU admissions.
- Economic Value: Projected 40% reduction in acute diabetes-related emergency hospitalizations.
- Patient Empowerment: Teaches patients how their specific physiology responds to foods & walks.

Conclusion & Challenge Audit

Happiest Health Digital Twin Challenge 2026

A complete, working, reproducible Healthcare Digital Twin.

- Satisfies 100% of Hackathon Challenge Requirements.
- Dual-Stream Data Fusion: Static EHR + Dynamic Wearables.
- Evolving Digital Twin Engine: State updates, memory buffer, what-if simulations.
- Proven ML Performance: ROC-AUC 0.989, Recall 91%, MAE 7.50 mg/dL, SHAP XAI.